

SHUVAM PAL



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unichronic

Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning, 3rd Year

Experience

Bentham AI

Dec 2025 – Feb 2026

Backend Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Built filing-workflow backend automation with browser execution, public-site **REST/API lookups**, **retries**, **session recovery**, validation checks, and operator checkpoints, reducing manual filing time by **85%**.
- Modeled API responses, screenshots, generated drafts, form state, and failure checkpoints as **recoverable workflow records** so broken **external-system steps** could be inspected and re-run safely.
- Implemented a draft-generation flow for company-name suggestions, object descriptions, and filing-field drafts while keeping generated outputs inspectable before deterministic backend execution.
- Shipped Dockerized **Node.js/Go backend services** on **Google Cloud Run** with **GitHub Actions** release workflows, improving deployment velocity by **70%**.

Google Summer of Code

May 2025 – Sept 2025

Mentee at Invesalio: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Integrated **Python workflows** for **95 anatomical brain regions**, wiring preprocessing, PyTorch/TorchScript and ONNX/TinyGrad execution paths, generated artifacts, and correctness checks into Invesalio.
- Reduced processing runtime by **87%** through **async/parallel execution** while debugging large data paths, orientation, label correctness, GPU/CPU execution, and reproducible output inspection.

Open Source

Contributions

- Contributed maintainer-reviewed patches across **Kuadrant MCP Gateway**, **Kyverno**, **Invesalio**, **Tiled**, **CRIU**, and **VideoLAN**, spanning backend failover behavior, parser robustness, security validation, compatibility fixes, and data/artifact correctness.

Projects

Seaweed Contest Platform | Go, React, Vite, PostgreSQL, Redis, Judge0, Docker, Kubernetes

GitHub

- Built a **Go/PostgreSQL platform** for **500+ concurrent users**, covering registration, problem pages, submissions, async judging, admin review, and live ranked leaderboards.
- Integrated Judge0 with **async Go dispatch**, **PostgreSQL result schemas**, **Redis runtime state**, Docker/Kubernetes deployment, and sub-**5s P95** verdict latency.
- Implemented **score aggregation**, **tie-breaking**, and leaderboard update logic so standings stayed consistent under repeated submissions and concurrent reads.

Postificus Content Distribution Platform | Go, Echo, PostgreSQL, Redis, RabbitMQ, Docker, Prometheus

GitHub

- Built a content-distribution **workflow backend** with **REST APIs**, auth/settings endpoints, **PostgreSQL persistence**, **Redis state/autosave**, **RabbitMQ workers**, S3-compatible storage, and live job/status tracking.
- Added **timeouts**, **retries**, **DLQs**, **circuit breakers**, **health checks**, and **Prometheus metrics** so long-running API/browser workflows stayed observable and recoverable under unreliable third-party integrations.
- Separated **workers**, **retry queues**, **failure states**, and service health checks across dependent services so external API/browser failures did not cascade through the platform.

Swish Support Workflow System | Python, Go, FastAPI, React, PostgreSQL, Redis, Langfuse

GitHub

- Built an **operational support workflow** that pulls order, kitchen, fleet, trust, and evidence context into assessment while **deterministic policy code** controls refunds, replacements, coupons, and escalations.
- Grounded support cases in **PostgreSQL/Redis state**, modeled active items, trust score, desired resolution, evidence strength, and **escalation path** instead of relying on prompt-only decisions.

Hyoka Evaluation/Replay Backend | Python, FastAPI, PostgreSQL, SQLAlchemy, OpenTelemetry, Docker

GitHub

- Built an evaluation/replay backend with **trace ingestion**, **validation runs**, replay workers, release gates, project-scoped API keys, **audit logs**, worker leases, and a Next.js dashboard.

Technical Skills

Languages: Go, Python, TypeScript, JavaScript

Frameworks: FastAPI, Node.js, React, Next.js, Echo, Gin, SQLAlchemy

Tools: PostgreSQL, Redis, RabbitMQ, Docker, Kubernetes, AWS, GCP Cloud Run, GitHub Actions, Prometheus, OpenTelemetry, REST APIs, HTTP